

DATA ANALYST – CANDIDATE PRESENTATION

Environmental Waste & Pollution Analysis Across UK Regions

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Presented to Southern Water – Wastewater Solutions Data Analyst Interview

CONTENTS – FROM DATA TO ACTION



THE “WHY”: PROJECT
CONTEXT AND BUSINESS
GOALS



THE “HOW”: MY
APPROACH & DATA
SOURCES



THE “WHAT”: KEYS
FINDINGS AND INSIGHTS



THE “TOOL”:
DASHBOARD
WALKTHROUGH



THE “NEXT”:
RECOMMENDATIONS,
PREDICTIONS, &
CONCLUSION

The Goal, The People & Why It Matterred



The Goal

- Analyse environmental incident data
- Understand trends over time
- Assess severity of incidents
- Identify risk concentration by region



Key Stakeholders



Environment Agency



Water Companies



Industrial Operators



Agricultural Sector



Local Authorities



Why It Matterred

- Supports regulatory compliance
- Reduces environmental risk
- Prioritises resources effectively

How I Analysed the Data

1

Data Cleaning

Standardised date formats, flagged unknown locations, checked for duplicates and inconsistencies

2

Structuring Data

Organised records by region, sub-region, date, impact category, and pollutant type

3

Pivot Tables

Used Excel pivot tables to count incidents by region, severity, and pollution medium

4

Pattern Analysis

Compared land, water, and air impact counts to identify which medium was most affected

5

Charts & Findings

Visualised regional breakdowns and pollutant types to tell a clear story from the data

What the Analysis Produced

Key measures from the full 36,000+ record dataset

17,053

Total Incidents

Pollution incidents recorded across all UK regions

94,969

Total Impact Score

Combined severity score across land, water and air impacts

5.6

Average Impact

Typical severity score per incident — our baseline measure

Southwest

Highest Risk Region

The region with the greatest concentration of high-severity incidents

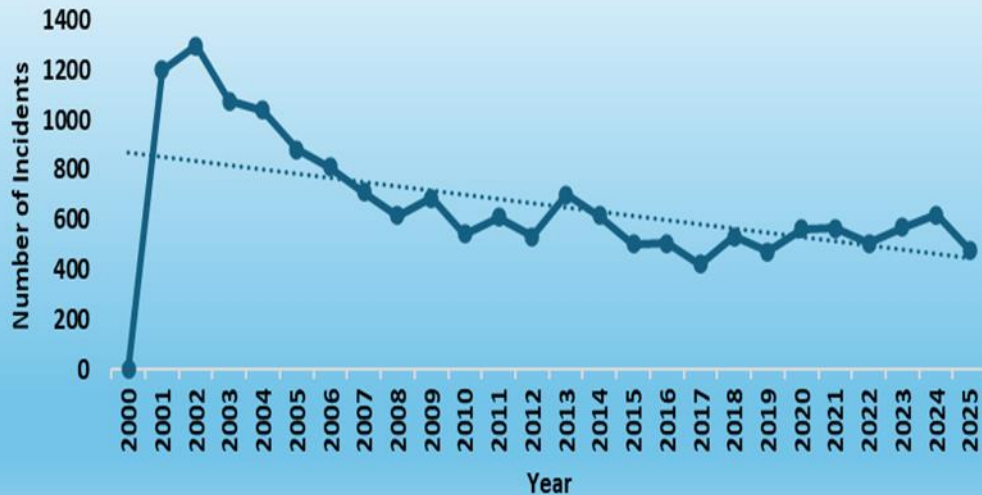
18,994

Projected Reduction

Estimated impact score saved by preventing 20% of high-risk incidents

Incident Volumes Are Falling Over Time

ANNUAL INCIDENT VOLUME (2000 - 20005)



Key Findings



Sharp drop after 2001

Incidents peaked at over 1,200 in 2001 then fell dramatically — suggesting early intervention and regulation had a real impact.



Steady long-term decline

The overall trend line shows a consistent downward trajectory over 25 years, indicating improving environmental management across the UK.



Recent uptick worth watching

Incident volumes show a slight rise from 2022 to 2025 — flagging this as an area that warrants closer monitoring going forward.

Summer Months Drive the Most Incidents

MONTHLY INCIDENT DISTRIBUTION



Key Findings



July is the peak month

July recorded the highest number of incidents at 2,010 — nearly double the quietest month. Summer activity and heat likely increase exposure and reporting.



December is the calmest

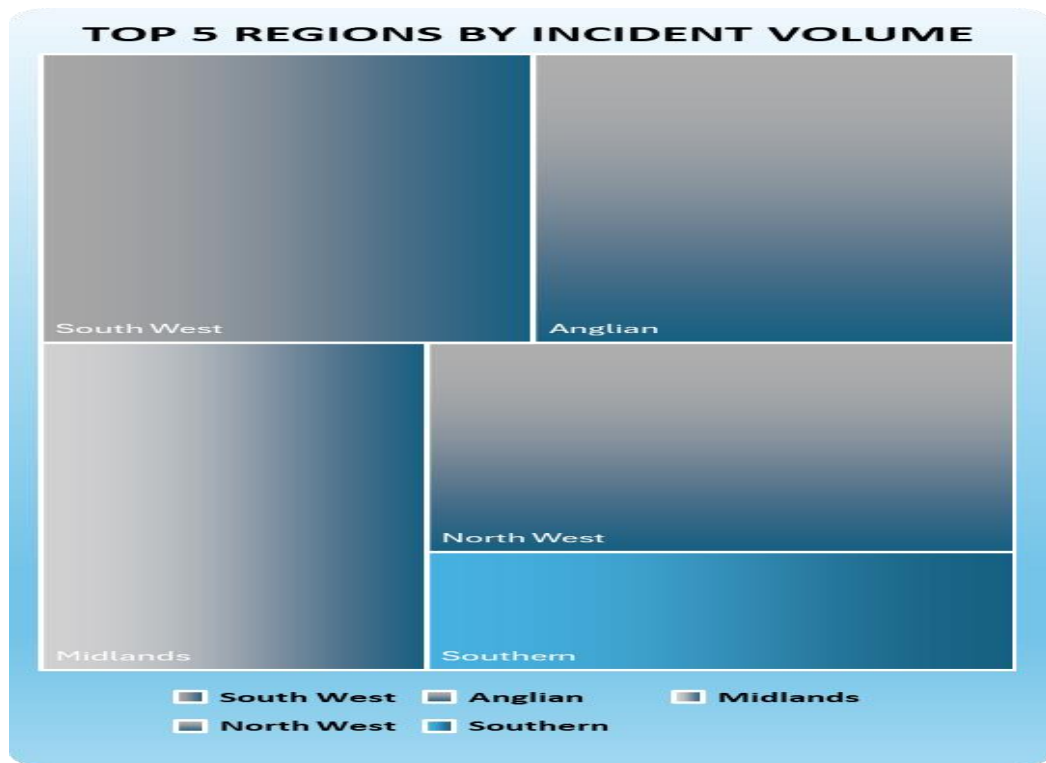
December had the fewest incidents at 909 — suggesting winter conditions reduce certain types of pollution activity.



Seasonal planning opportunity

The clear June–August peak gives environmental teams advance warning to increase monitoring capacity and field presence during high-risk months.

Southwest Is the Highest Risk Region



Key Findings

Southwest leads in volume

The Southwest accounts for the largest share of incidents — confirming it as the highest risk region and the priority area for intervention.

Anglian region is second

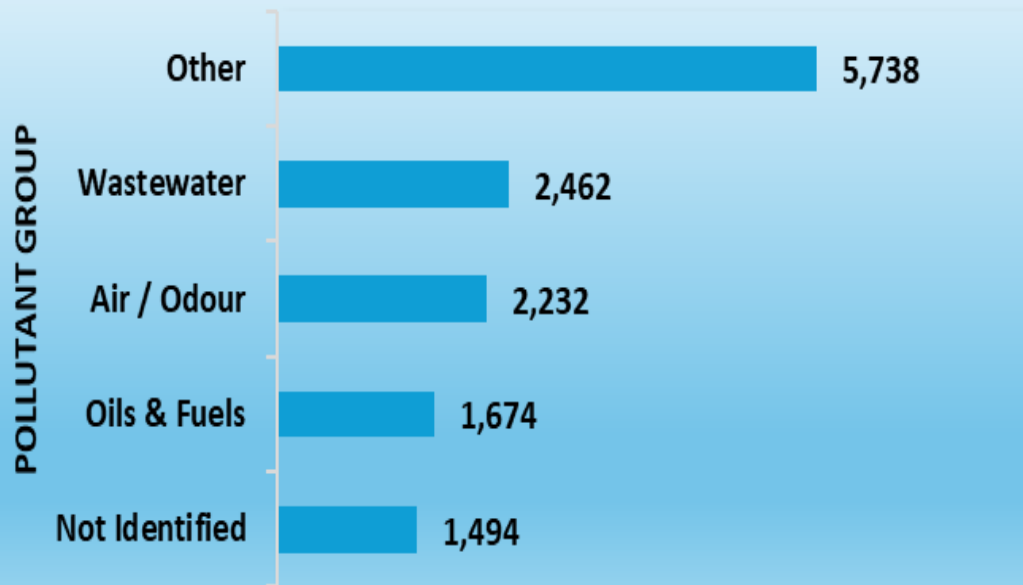
Anglian recorded the second highest volume — both coastal regions, suggesting proximity to water bodies may increase incident risk.

Resources should follow risk

With Southwest and Anglian dominating, targeted deployment of monitoring teams and response resources to these regions would have the greatest impact.

Wastewater & Air Pollution Are the Biggest Threats

TOP 5 POLLUTANT TYPES



Key Findings



Wastewater is a major concern

With 2,462 incidents, wastewater is the second largest pollutant group — directly relevant to Southern Water's core responsibility of managing sewage and effluent.



Air and odour incidents significant

2,232 air and odour incidents highlight that pollution impacts go beyond water — affecting the quality of life for nearby communities.



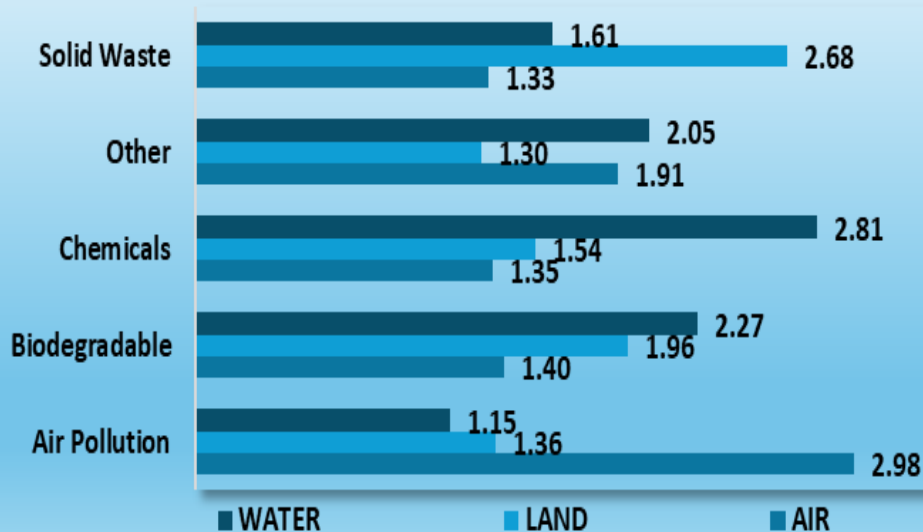
5,738 unclassified incidents

The largest group is 'Other' — suggesting a gap in reporting standards that makes it harder to target the true root causes of pollution.

Water Bears the Highest Impact Across All Pollutant Types

IMPACT SEVERITY BY POLLUTANT TYPE

POLLUTANT TYPE GROUP



Key Findings



Water is the most impacted medium overall

Across all pollutant types, water has the highest average impact score of 2.35 — compared to land at 1.64 and air at 1.57. Water consistently bears the greatest environmental burden.



Agricultural pollution hits water hardest

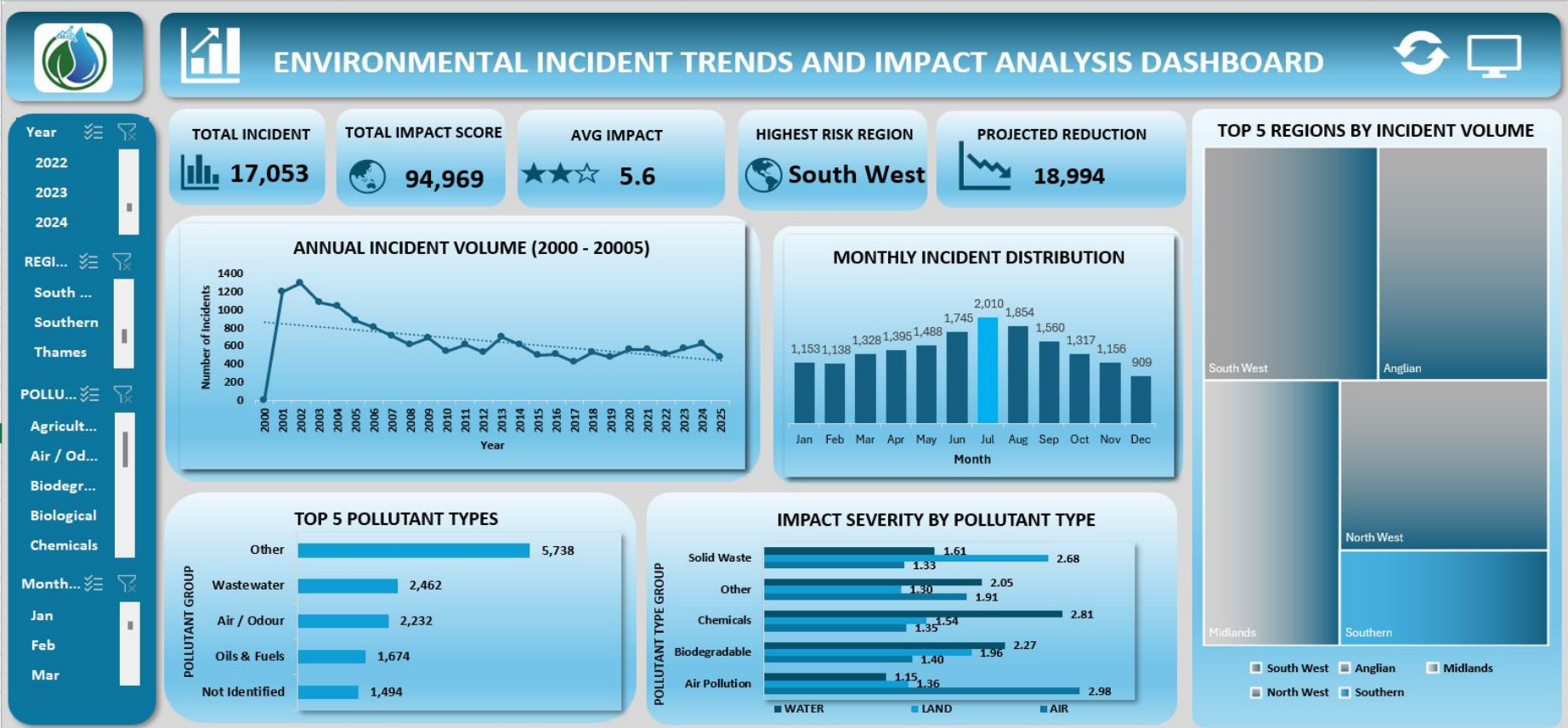
Agricultural sources score 3.08 for water impact — the single highest score in the entire dataset. Farm run-off is the most critical threat to water quality.



Wastewater and unknown sources also critical

Wastewater scores 3.01 and unknown sources 3.03 for water impact — highlighting both a treatment gap and a reporting gap that need urgent attention.

DASHBOARD OVERVIEW



Key Patterns & Insights



Water Quality Deterioration Near Waste Sites

Regions with a high concentration of landfill and industrial waste sites consistently showed lower water quality index scores, suggesting a direct link between waste proximity and contamination risk.



Industrial Activity as a Key Driver

Areas with higher industrial activity recorded significantly elevated pollutant concentrations. This pattern held across multiple pollutant types, pointing to industrial operations as a primary risk factor.



Regional Disparities Are Significant

The gap between the highest and lowest-risk regions was considerable — suggesting that targeted, region-specific interventions would be far more effective than blanket national policies.



How I Validated the Findings

Cross-referenced results across multiple data sources, checked for data entry errors, and tested whether patterns remained consistent across different time periods to ensure reliability.

What I Recommended & Why It Mattered

My Recommendations

-  Prioritise monitoring and intervention in Southwest and Anglian— the highest-risk regions identified in the analysis
-  Track sewage and wastewater incidents closely. With over 2,400 incidents, they directly affect downstream water quality, which is core to Southern Water operations
-  Engage with agricultural operators. Farm run-off scored the highest water impact of 3.08 in the entire dataset, making it the most critical threat to water quality
-  Build on this analysis over time. The 25-year trend shows improvement but the recent uptick from 2022 warrants continuous monitoring, not a one-off snapshot

Impact of This Work

The analysis provided stakeholders with a clear regional risk picture for the first time, enabling evidence-based prioritisation of resources and monitoring efforts.

Future Improvements

- Extend to multi-year dataset for trend analysis
- Add geospatial maps using coordinates already in the data
- Build automated dashboard for live incident tracking

Why This Experience Is Relevant to Southern Water

My Experience

Analysing pollution levels across UK regions



Working with complex, multi-source environmental data



Translating data into clear insights for non-technical stakeholders



Identifying high-risk areas and recommending targeted action



Southern Water Need

Southern Water monitors wastewater quality and environmental impact across its network

Southern Water combines data from treatment plants, sensors, and compliance reports

Southern Water teams need clear reporting to support operational and regulatory decisions

Southern Water prioritises interventions based on risk, compliance, and environmental impact

Thank You

I'm excited to bring my environmental data experience to the Wastewater Solutions team at Southern Water.

17,053

Incidents
Analysed

9

UK Regions
Covered

2

Datasets
Combined

Excel

Primary
Tool Used